

RIDHI SRIKANTH

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SUMMARY

UC San Diego Computer Engineering Honors student (3.99 GPA) specializing in Systems Research and Compiler Design. Currently modifying CPython internals to develop safety-focused Domain Specific Languages (DSLs) for scientific workflows. Aiming to leverage low-level systems expertise to build robust, reproducible computational infrastructure.

EDUCATION

University of California, San Diego | BS, Computer Engineering | **GPA:** 3.99 | **Graduation:** June 2027

- **CSE Dept. Honors Candidate:** Faculty-advised research track culminating in an Honors thesis.
- **Awards:** Eta Kappa Nu (HKN) IEEE Honor Society; Provost Honors (7 consecutive quarters), 1st place UCSD WIC programming competition Fall 2023
- **Selected Coursework:** Advanced DSA, Operating Systems, Parallel Programming, Computer Architecture, Machine learning

SKILLS

- **Programming Languages:** Java, Python, C, C++, TypeScript, JavaScript, HTML, SQL, CSS, Shell, Rust
- **Technologies and Expertise:** React, Flask, Django, Agile, CI/CD, Git, Linux, Embedded Systems, Firebase

LEADERSHIP

- IEEE UCSD Chapter Technical Chair: Led 5+ technical workshops for 100+ students, including live-coding sessions on React and recommendation systems, fostering hands-on learning for peers

RESEARCH EXPERIENCE

Research Lead - UC San Diego CSE Department - (Advisor: Professor Michael Coblenz)

Enhancing Shell Scripting Workflows by Modifying Python Compiler in C

June 2025 - Present

- Research goal: To improve the safety and usability of shell scripting in scientific workflows
- Leading an independent research project; Performed thematic and static analysis of shell scripts from scientific workflows
- Leveraging findings to design a Python-based Domain Specific Language (DSL) for safer and accessible shell scripting
- Working across CPython's compiler pipeline: modifying PEG grammar, parser logic in C, AST generation, bytecode emission, and stack behavior to integrate new language features.
- Writing C functions and updating VM evaluation logic to implement the DSL's execution semantics.
- Using GDB and backward-tracing to debug low-level issues, including stack inconsistencies and buffer overflows.

Research Assistant - Progsys Lab UCSD

Scientific Workflow Reproducibility Study

Feb 2025 - July 2025

- Collaborated with a research team on a corpus study of 500+ scientific programs
- Analyzed selected programs for runtime errors, dependency issues, and execution failures affecting software reproducibility
- Co-authored a research paper; Manuscript under revision

TECHNICAL EXPERIENCE

ECE 35 (Analog Design) Course Tutor

Sep 2025 - Present

- Managing support for 400+ students, providing tutoring and hands-on lab guidance on real device circuit implementation
- Collaborating in weekly meetings to refine instructional strategies and responsible for grading quizzes and final exam questions of fundamental circuit theory concepts.

Full-stack Developer - ElectroTriton With Professor Curt Schurgers | [ElectroTriton](#)

Aug 2025 - Present

- Developing features in ElectroTriton, a Django and MySQL homework management web app used for Engineering courses
- Implemented course cloning functionality which accomplished a ~71% reduction in course setup time (35→10 min)
- Automated student list updates which reduced roster management time by ~78% (45 → 10 min).

Full-stack Developer - LabStream With Professor Saharnaz Baghdadchi | [LabStream](#)

Oct 2024 - June 2025

- Developed, with team of 8, a scalable web application enabling real-time remote access to electrical engineering labs
- Built a real-time slot booking system that reduced scheduling time for lab sessions by ~65% (multiple emails → 5 mins)
- Implemented features including lab session creation, active user session management, and role-based access control.
- Evaluated and integrated Firebase for data storage, CRUD operations, authentication; developed reusable React components

PROJECTS

Hardware Developer | [Hardware Blog](#): Designed and prototyped 5+ embedded systems with C++ firmware and custom PCBs.